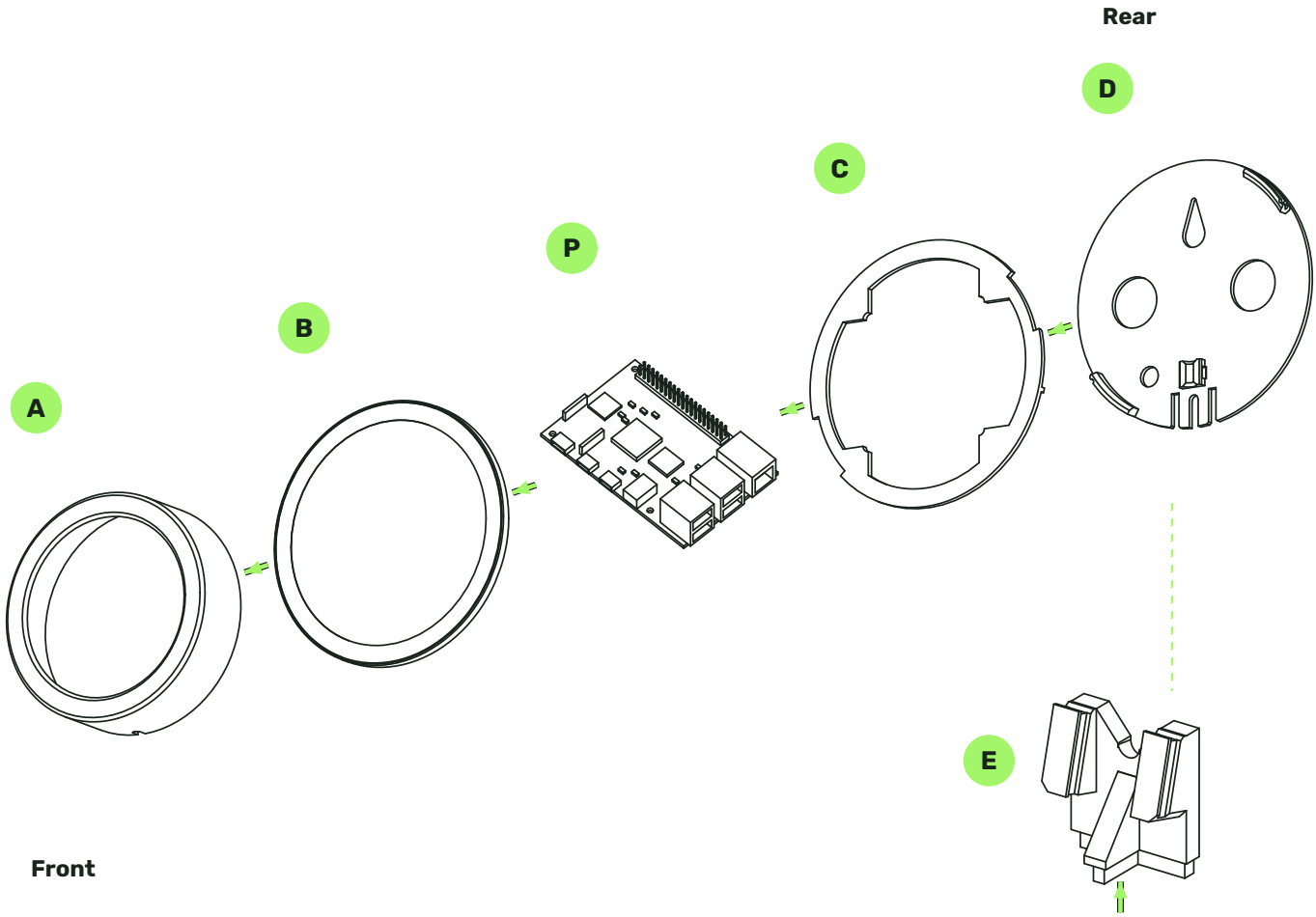
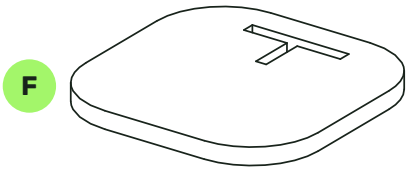
Assembly Manual

Build Your AroundOS Computer

Five printed parts. One round display. A Raspberry Pi desktop.



A Front Case
B Display Module
P Raspberry Pi 4
C Holder Plate
D Rear Plate
E Stand Connector
F Desk Base



Read Before Building

This guide follows the supplied V4R2 enclosure and SKU 24369 screen CAD. The Pi mounts to the screen using its supplied spacers and screws. Drawings show component geometry; separated positions and arrows explain the sequence and are not fit measurements.

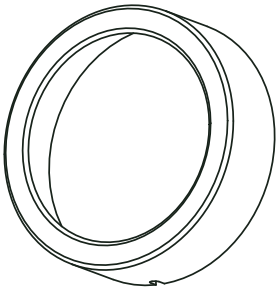
Physical fit, latch engagement and stand stability still require a build check. No assembly or powered hardware test was performed for this guide.

Parts + Equipment

Lay Out the Build

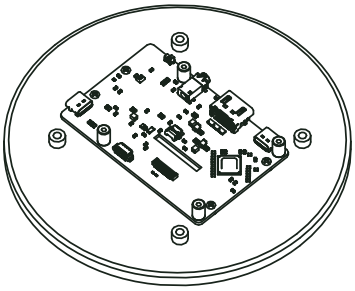
Match the letters throughout the guide. Print A, C, D, E and F; purchase B as a complete module.

A Front Case x1



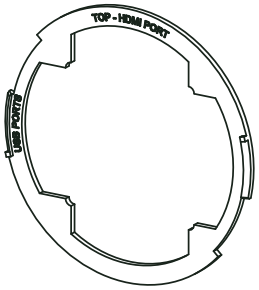
The deep outer shell and front opening.

B Display Module x1



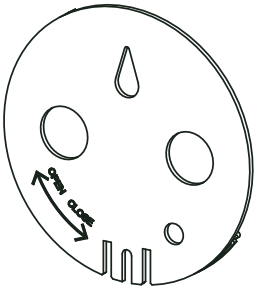
Purchased Waveshare screen; keep its factory assembly intact.

C Holder Plate x1



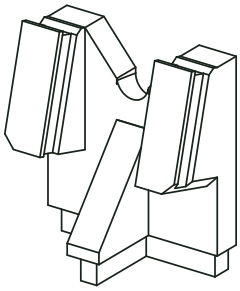
The open retaining ring behind the display.

D Rear Plate x1



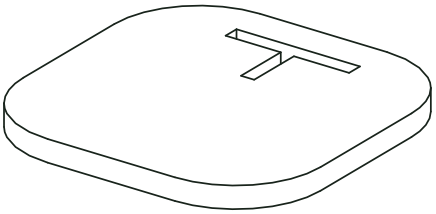
The rear cover with cable opening and mount features.

E Stand Connector x1



The small link between rear cover and base.

F Desk Base x1



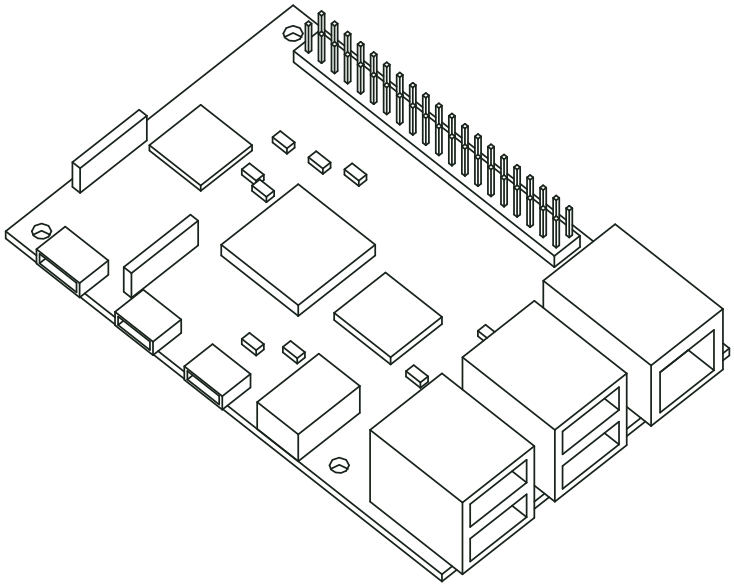
The flat foot of the desk stand.

Also Gather

Raspberry Pi 4 Model B; four supplied spacers and matching screws; microSD card; Pi USB-C supply; short micro-HDMI to HDMI lead; USB-A to USB-C data cable; keyboard; small screwdriver; clean soft cloth. Choose a cooling arrangement that clears the rear cover.

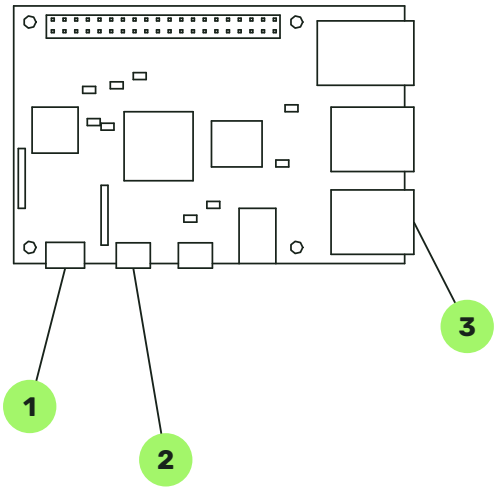
Know Your Raspberry Pi

A wire-style board illustration shows the connections used in this build. Match the socket shapes on your board.



Perspective View

Top View



1 USB-C Power Input

Connect the Pi power supply here. Power enters the computer at the Raspberry Pi.

2 Micro-HDMI 0

Use the HDMI socket nearest the USB-C power input. Connect it to the screen HDMI port.

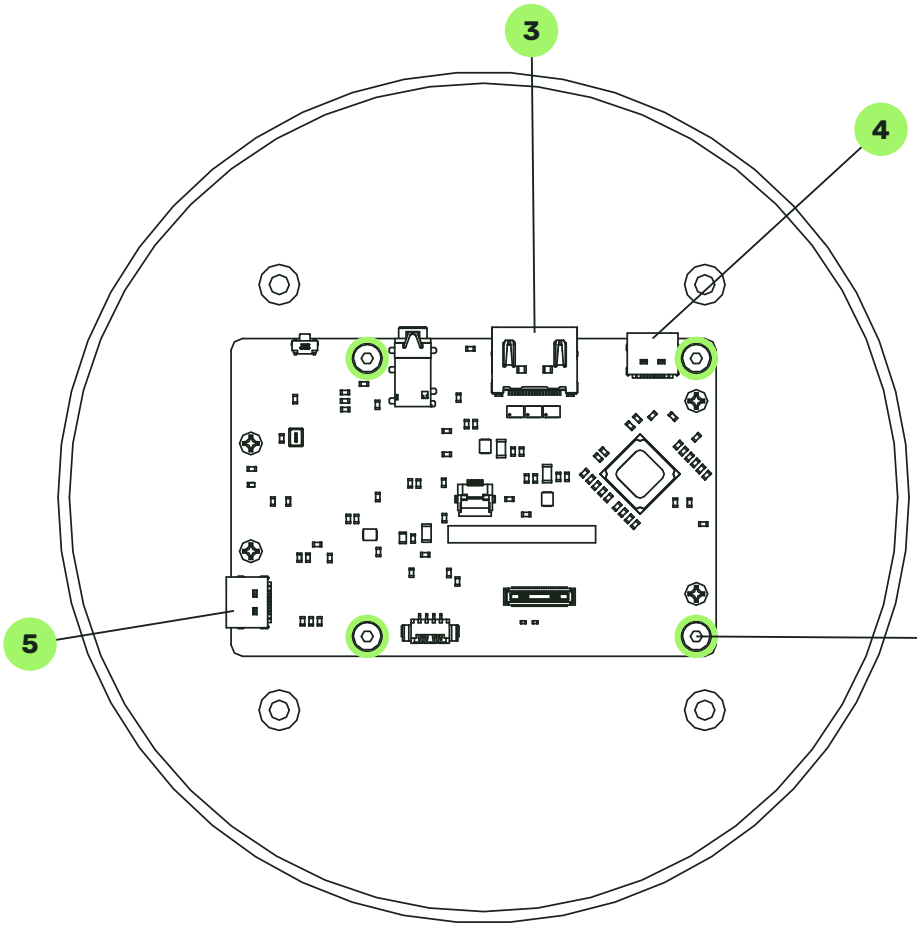
3 USB-A for Touch

Connect a USB-A data port to the screen USB-C Touch socket. HDMI alone does not carry touch.

Four board mounting holes align with the screen's Pi mounting points. Components, connector shells and pin details are simplified; this illustration is not a fabrication drawing.

Identify the Screen Connections

Rear view, oriented like your supplied Waveshare photo. Numbers 3, 4 and 5 match the labels in that reference.



Four Pi Mounting Points

Highlighted rings. Use the supplied spacers here.

3 HDMI Input

Connect to Raspberry Pi micro-HDMI 0 for video.

5 USB-C Touch

Connect to a Pi USB-A data port for touch and the USB supply path.

4 Unused Port 4

Leave this socket unconnected. Connect the power supply only to the Raspberry Pi; use screen port 5 for USB touch.

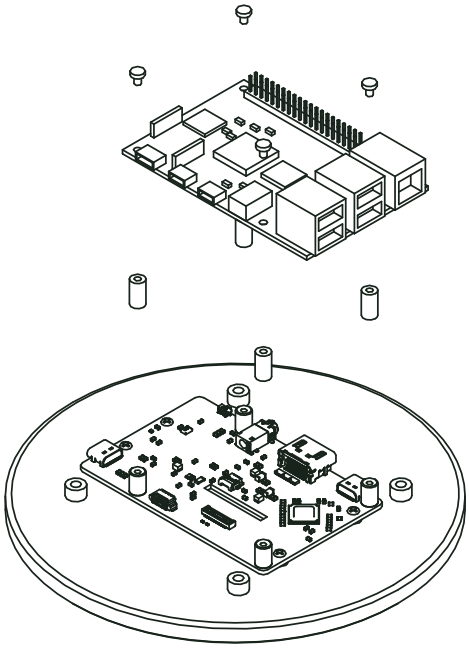
Other photo labels: 1 backlight on/off; 2 earphone jack; 6 speaker header. The four separate bosses on the outer round plate are not the Pi mounting points.

Assembly 01

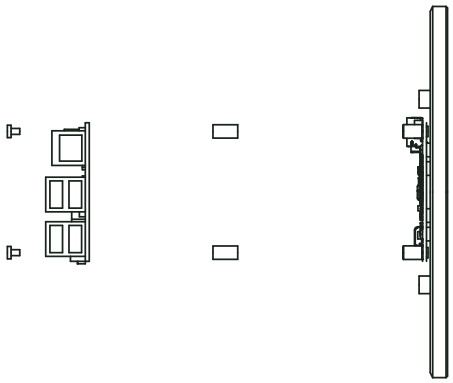
Attach the Pi to the Screen

Use the screen's supplied spacers and screws. Work with power disconnected and the screen supported face down on a clean, soft surface.

4 Screws / Pi / 4 Spacers / Screen



Side Alignment



Exploded views: all four Pi holes align with the four spacers. Separation is exaggerated.

1. Prepare Raspberry Pi OS 64-bit Desktop on a microSD card and insert it into the Pi. Keep a keyboard or SSH access available for the bench test.
2. Identify the four Pi mounting points on the screen PCB shown on page 4. Use these, not the outer round-plate bosses. Leave the screen factory screws in place.
3. Fit the four supplied spacers to those mounting points. Start them by hand and seat them evenly without forcing the threads.
4. Lower the Pi onto the spacers with its components facing away from the screen. Keep its USB-A and Ethernet bank toward the right and align all four holes.
5. Fit the supplied screws through the Pi into the spacers. Start all four, then tighten gently in a diagonal sequence until the board is secure and flat.

Check Before Continuing

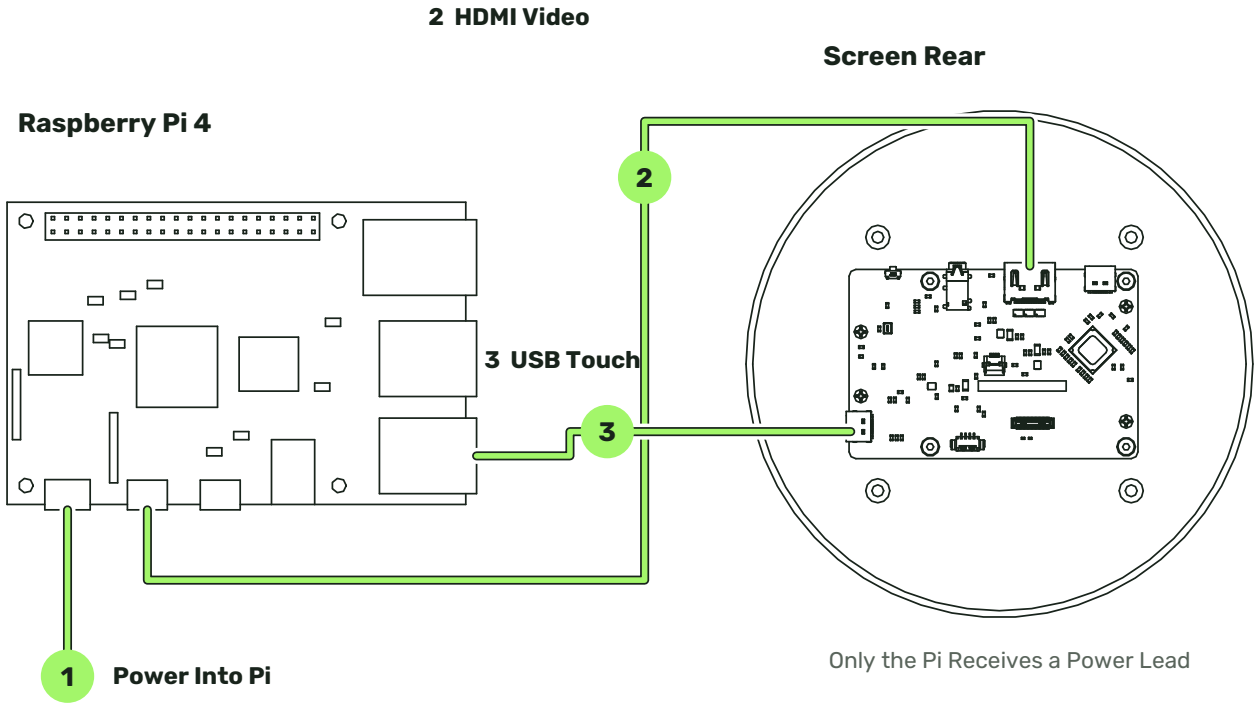
All four holes line up, every spacer is seated, and the Pi is held without bending or touching the display electronics.

Spacer height and screw length are not specified here; use the supplied matching set. Exploded gaps are enlarged for clarity.

Assembly 02

Wire and Test on the Bench

Connect and test the Pi while it is mounted to the screen, before starting the enclosure. Power connects only to the Raspberry Pi.



Cable routing is schematic. 1: Pi power. 2: Pi HDMI0 to screen 3. 3: Pi USB-A to screen 5.

1. Connect Pi micro-HDMI port HDMI0 to the screen's standard HDMI input. Use a micro-HDMI-to-HDMI cable or the matching adapter.
2. Connect a Pi USB-A port to the screen's USB-C Touch port with a data-capable cable. HDMI alone does not provide touch.
3. Connect the power supply only to the Pi USB-C power input, after HDMI and USB touch are connected. Leave screen port 4 unused.
4. Boot, confirm an image and test touch at 1080 x 1080, approximately 60 Hz. Shut down and unplug the Pi power supply; keep HDMI and USB touch connected for enclosure assembly.

Check Before Continuing

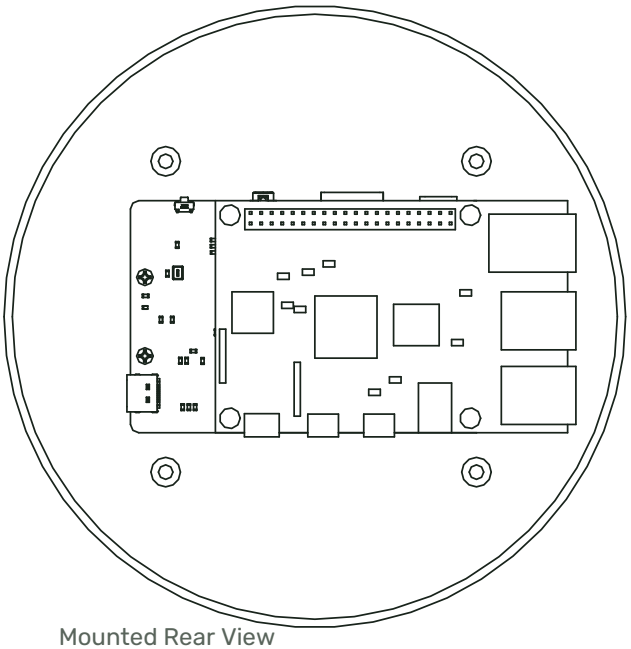
The screen displays the desktop and touch follows your finger before any cover is fitted.

Power Supply > Raspberry Pi USB-C. Video > screen HDMI port 3. Touch > screen USB-C port 5 to Pi USB-A.

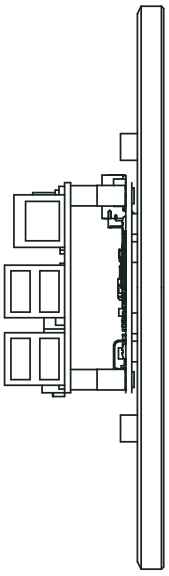
Assembly 03

Check the Connected Assembly

The Pi is now attached and connected to the screen. Finish these checks before preparing or fitting any enclosure parts.



Mounted Rear View



Board Separation

Connected screen/Pi unit before case assembly. HDMI and USB touch leads are omitted for clarity.

1. With the Pi power supply unplugged, check both ends of HDMI and USB touch are fully seated. Keep these two leads connected through the case assembly steps.
2. Inspect the gap between the boards. The Pi must sit flat on all four spacers with no loose screws or contact against the display electronics.
3. Arrange the two interconnects around the board edges, clear of GPIO pins. Leave slack for fitting the holder and rear plate without pulling on either socket.
4. Lift the complete unit by supporting the screen rim, not the Pi or cables. Set it aside on a clean, soft surface before cleaning and checking the printed parts.

Check Before Continuing

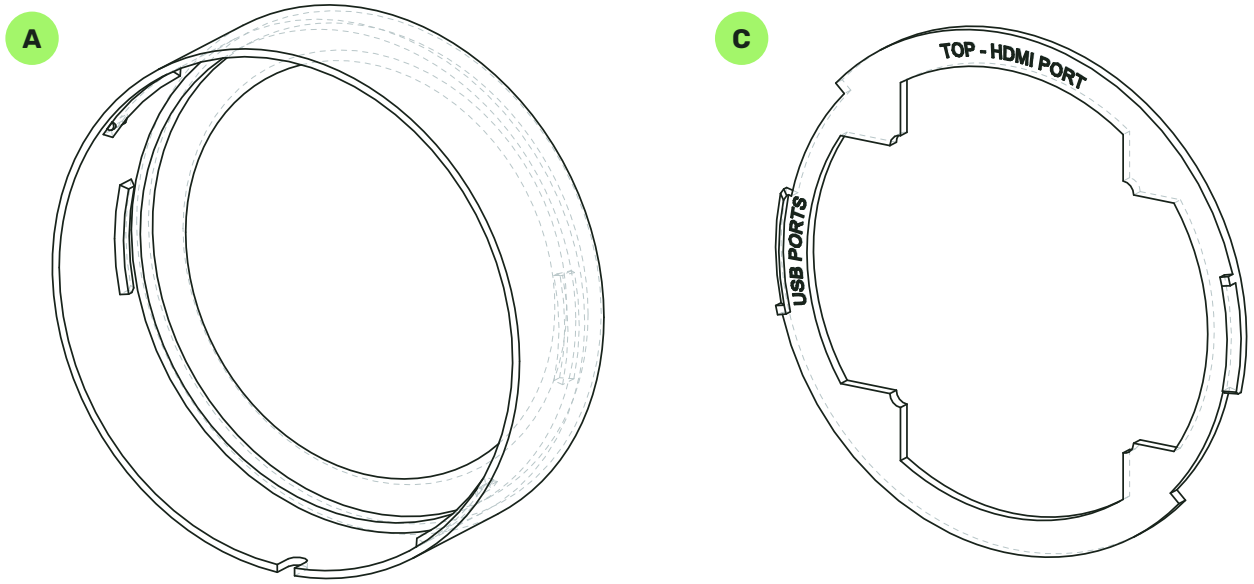
The screen and Pi are attached, HDMI and touch have passed the bench test, and the power supply is unplugged.

The CAD views illustrate alignment. Actual spacer height, cooler, cable plugs, enclosure clearance and cooling require a physical check.

Assembly 04

Prepare the Printed Parts

Set the assembled and tested screen/Pi unit aside while preparing the five printed parts. Keep the electronics away from print-cleanup dust.



Rear-facing CAD views. Prepare the prints with the connected electronics set aside.

1. Import the five STL files at 100% scale in millimetres. Centre each part on the build plate and check that it sits on the bed; several supplied files have offset coordinates.
2. The folder's source notes specify support-free printing in the supplied STL orientations. Inspect the slicer preview, particularly clips and bridges; STEP exports need orientation before slicing.
3. Remove brim, strings and loose material. Smooth only printing defects on mating edges; preserve the designed lips, tabs and latch shapes.
4. Dry-fit the stand connector to the base and inspect the case's retaining features. Do not force a tight fit against the glass. Correct the print before installing the screen.

Check Before Continuing

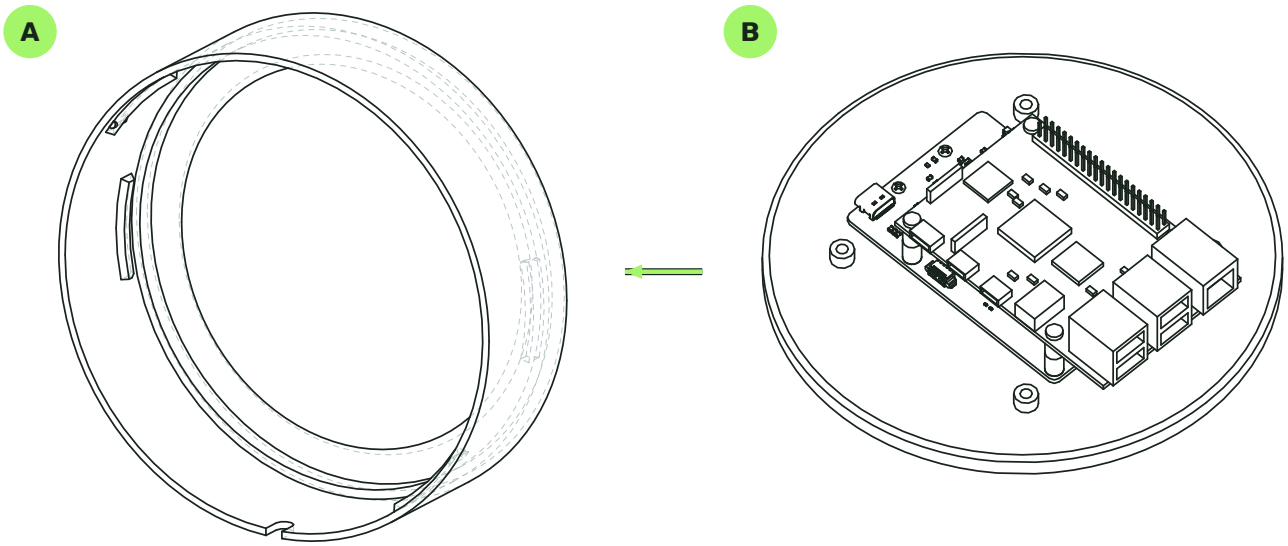
All five parts are clean, flat where intended, and free of cracked or obstructed retaining features.

No universal material, layer-height or infill profile is supplied. Print shrinkage and tolerances must be checked on your own printer.

Assembly 05

Seat the Connected Unit in the Case

Install the screen with the Pi already mounted and HDMI and USB touch already connected. Keep the Pi power supply unplugged.



Open Rear of Front Case

Pi Already Attached and Connected

Lower the connected screen/Pi unit into A. HDMI and USB touch leads are omitted from this view for clarity.

1. Place front case A with its front rim supported on a clean, soft surface and its large rear opening facing up.
2. Hold the connected screen/Pi unit by the screen rim. Orient the glass toward the front opening and keep all cable loops clear of the seating edge.
3. Lower the complete unit into the case from the rear. Keep it square so the screen outer edge settles evenly against its internal seating lip.
4. Check the screen is centred and the Pi, cooler and connected plugs clear the shell. Stop if anything binds; lift the unit out and inspect the fit without forcing it.

Check Before Continuing

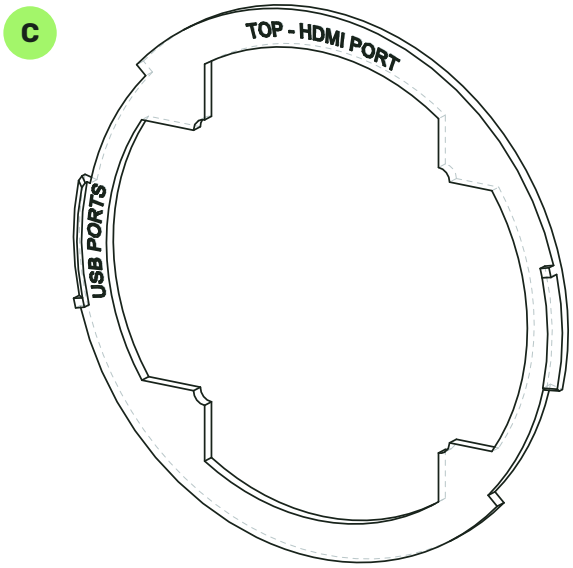
The connected unit sits evenly in the case; the glass, Pi and cables are free from pressure.

The exploded spacing is illustrative. This guide does not establish a measured snap-fit clearance or a tested insertion force.

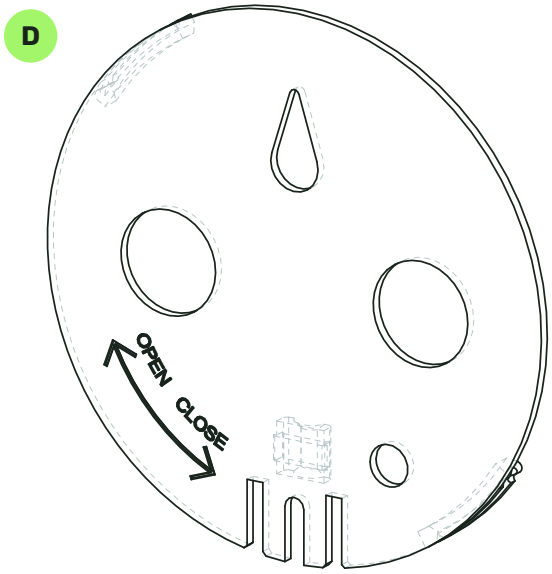
Assembly 06

Fit the Screen Holder

Fit holder plate C around the already mounted Pi and connected leads. The holder retains the screen within the case.



Holder Clears the Mounted Pi



Route the Pi Power Lead

Fit the holder around the mounted Pi and connected leads. The wiring map is on page 6.

1. Guide holder plate C around the Pi and its connected leads. Align TOP - HDMI PORT and USB PORTS with the screen connectors, then match the holder tabs to the case.
2. Keep the ring clear of the Pi, components and cable plugs. Engage the matching case features evenly by hand; do not use the board or glass as a pressure point.
3. Check the holder is seated and the screen remains located. Confirm HDMI and USB touch are still fully connected and no cable is caught under the ring.
4. Offer up rear plate D and route the Pi power lead through its cable exit. Keep the power supply unplugged and check the cover clears the cooler, plugs and cable bends.

Check Before Continuing

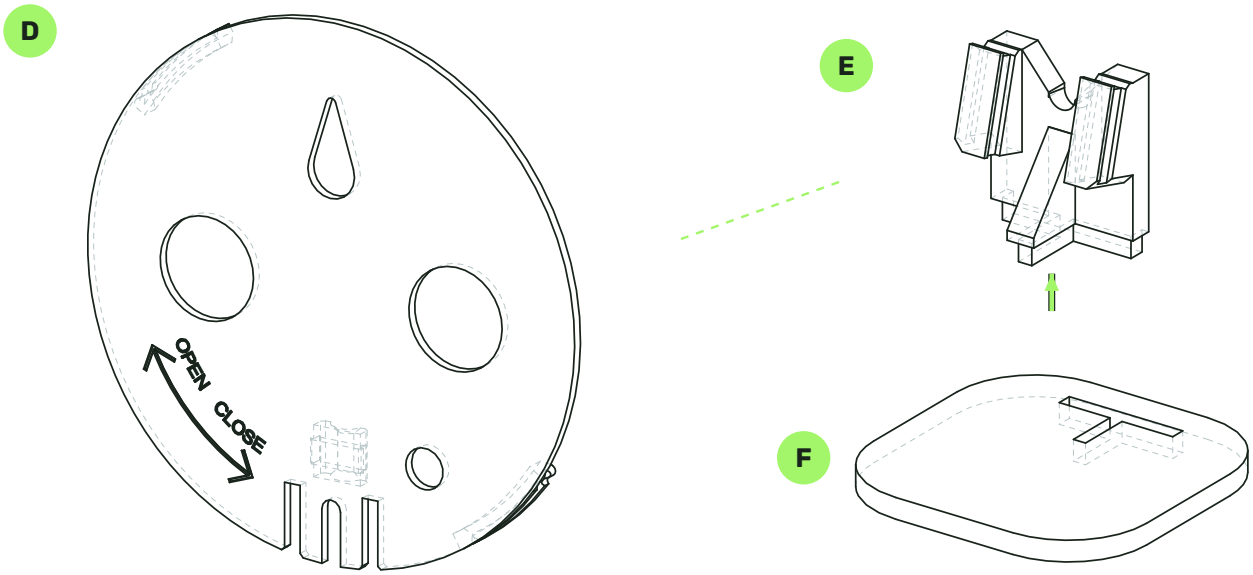
The holder retains the screen, both signal leads remain connected, and the Pi power lead exits freely.

The printed enclosure uses slide/click features. The separate Pi-to-screen attachment uses the supplied spacers and screws.

Assembly 07

Close the Back and Fit the Stand

Test the stand's two interfaces before loading them with the display. Keep the screen supported while engaging the rear cover.



Separate CAD views of the rear cover and stand. Mate the shaped interfaces on the actual parts.

1. After the Pi is mounted and all leads are clear, align rear plate D with the case. Keep the stand attachment downward and check the Pi, cooler and plugs clear the inside of the cover.
2. Engage the rear cover tabs and turn gently in the moulded CLOSE direction. Use the OPEN marking when removing it. Watch the perimeter so no cable or connector is pinched.
3. Align connector E with the receiving feature in base F and seat the matching tongue/slot connection. Check the connection holds before attaching the display.
4. Support the display and mate the connector's other end with the rear plate mount. Follow the direction allowed by the matching features; do not twist or hammer them together.
5. Place the assembly on a level desk. Support it as you check that the base sits flat, the display is retained and the cable slack does not pull the stand over.

Check Before Continuing

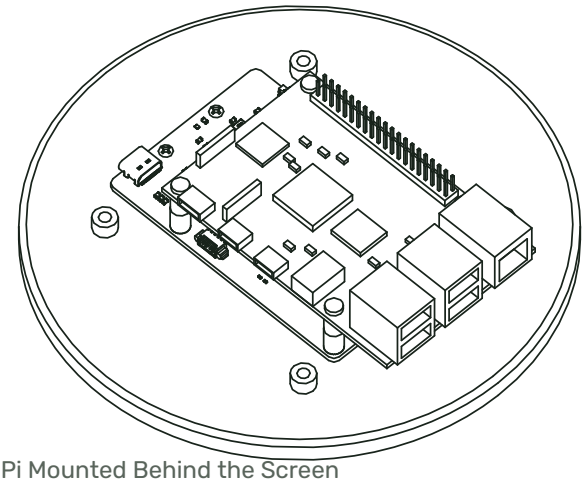
The cover is even, both stand joints hold, and the assembly remains stable when you gently use the touchscreen.

This guide covers the desk build. Wall anchors, screw selection and wall load ratings have not been specified or verified.

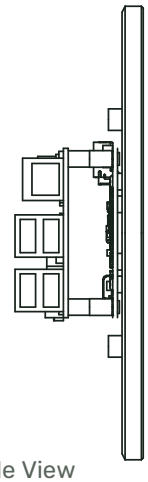
Assembly 08

Connect and Start AroundOS

The Pi is mounted behind the screen. Its USB-C socket receives power; HDMI and USB touch link it to the Waveshare display.



Pi Mounted Behind the Screen



Side View

From the Repository on the Pi

python3 scripts/round-desktop.py install

1. Check the Pi is retained on all four spacers. Confirm HDMI and USB touch are connected, the power lead exits freely, and the cooler and vents are unobstructed.
2. Reconnect power and boot Raspberry Pi OS Desktop. Confirm display and touch still work after closing the enclosure.
3. Check that the active display is HDMI-A-1 at 1080 x 1080, approximately 60 Hz, with scale 1. Keep a keyboard or SSH fallback for setup.
4. Follow the repository's Getting started instructions. Run the desktop installer as the normal desktop user inside the active desktop session, then install optional applications as needed.
5. Open the launcher and a few apps. Check touch at several visible positions, cable strain, stand stability and unobstructed Pi ventilation.

Check Before Continuing

AroundOS opens on the round display, touch works, and the assembled unit remains stable in normal use.

AroundOS is an experimental desktop customization, not a flashable OS image. Fresh-device installation is not yet fully validated.

Finish + Fault Finding

Check the Complete Build

A successful build has a retained screen, clear cable paths, a stable stand and a working desktop.

- [ ] Screen sits evenly; glass and electronics are not under pressure.
- [ ] Holder and rear cover are fully engaged with no cracked clips.
- [ ] No cable is caught in a seam or pulled tight against a connector.
- [ ] Both stand joints hold and the base stays flat during touch use.
- [ ] HDMI image and USB touch work after the cover is closed.
- [ ] Pi is secure on four spacers; its cooler and airflow remain clear.

If Something Does Not Work

No Picture	Recheck Pi power, Pi boot, HDMI0 and the USB touch cable. Test with the rear cover open.
Picture, No Touch	Use the screen Touch socket and a USB data cable. Confirm the Pi detects USB touch.
Flicker or Resets	Check the Pi power supply and USB touch cable. Repeat the bench test before closing the cover. Power connects only to the Pi.
Screen or Cover Binds	Unplug power. Remove the affected part; inspect scale, print defects, alignment and cable clearance.
Desktop Setup Fails	Use the repository Getting started and troubleshooting sections. Check output name, resolution and scale.

To Reopen

Shut down, disconnect power and support the display. Inspect which retaining feature releases before moving a cover or stand joint. Release it gently; avoid prying against the glass or pulling on cables.

Record your physical fit checks before treating this guide as validated for repeat builds.

References + CAD Notes

What This Guide Is Based On

Enclosure and screen views use supplied STEP geometry. The Pi and mounting hardware are simplified line illustrations.

Model Observations

The front case spans about 162 mm across and 49.3 mm in depth. The holder spans about 156.8 mm. The screen STEP assembly spans about 150 mm across and 18.1 mm including modelled rear components. These are CAD bounding extents, not manufacturing tolerances or fit certification.

View Conventions

Dark lines show visible edges; pale dashed lines show hidden edges. Neon green (#A3F66A) marks connections and actions. The Pi is a simplified illustration based on its mechanical outline, not supplied production CAD. Spacer height, exploded gaps and cable routes are illustrative. Check actual hardware and cover clearance.

Sources and Further Instructions

1 Enclosure Design and Original Instructions

JN-Husch (@JNHusch), Case for round Waveshare 5in Display.

https://www.printables.com/model/847901-case-for-round-waveshare-5in-display

2 Screen Connections and Electrical Information

Waveshare, 5inch 1080x1080 LCD wiki; SKU 24369.

https://www.waveshare.com/wiki/5inch_1080x1080_LCD

3 Raspberry Pi 4 Board and Mounting Pattern

Raspberry Pi official mechanical drawing: 85 x 56 mm; 58 x 49 mm holes.

https://datasheets.raspberrypi.com/rpi4/raspberry-pi-4-mechanical-drawing.pdf

4 AroundOS Installation and Current Limitations

Repository README and docs/round-desktop/README.md.

https://github.com/SalvSith/AroundOS

Attribution and Review Status

Enclosure design and derived enclosure illustrations: CC BY-NC-SA 4.0, credited to JN-Husch. Preserve the parent folder LICENSE.txt. New manual text is offered under CC BY-NC-SA 4.0. Waveshare screen geometry remains credited to Waveshare; no separate redistribution license was supplied for that model. No endorsement is implied.

Guide authored 7 November 2026. The source enclosure page was unavailable during this pass; print/assembly notes use the attributed folder README and supplied exploded view. Port labels follow the supplied Waveshare details-17 photo. The user confirmed the supplied mounting hardware. Screen STEP read directly; EXE not executed. Physical fit and powered assembly remain unverified.